

DFA INT

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
char stat_table[3][3][10] = {  
    {"stat", "digit", "letter"},  
    {"start", "INT", "error"},  
    {"INT", "INT", "error"}};
```

```
int main() {
```

```
    int i, len, row, column, r, c, flag = 0, error = 0;
```

```
    char input[20], column_stat[10], next_stat[10], current_stat[10];
```

```
    char ch;
```

```
    clrscr(); // Clear the screen
```

```
    printf("Enter Integer:");
```

```
    gets(input);
```

```
    flag = 0;
```

```
    error = 0;
```

```
    len = strlen(input);
```

```
    printf("OUTPUT:\n");
```

```
    strcpy(current_stat, "start");
```

```

for (i = 0; i < len; i++) {
    for (r = 1; r < 3; r++) {
        if (strcmp(stat_table[r][0], current_stat) == 0) {
            row = r;
            break;
        }
    }

    ch = input[i];

    if (!isdigit(ch)) {
        strcpy(next_stat, "error");
        error++;
    } else {
        strcpy(column_stat, "digit");
    }

    if (error == 0) {
        for (c = 1; c < 3; c++) {
            if (strcmp(stat_table[0][c], column_stat) == 0) {
                column = c;
                break;
            }
        }

        strcpy(next_stat, stat_table[row][column]);
    }
}

```

```

        printf("%s %c %s\n", current_stat, ch, next_stat);
    } else {
        printf("%s %c %s\n", current_stat, ch, next_stat);
        printf("ERROR: Invalid Input\n");
        break;
    }

    strcpy(current_stat, next_stat);
}

if (error == 0 && strcmp(current_stat, "INT") == 0) {
    printf("Valid Integer\n");
} else {
    printf("Invalid Integer\n");
}

getch(); // Wait for a key press
return 0;
}

```